INCH-POUND
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17 July 1989
SUPERSEDING
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MILITARY SPECIFICATION

VIBRATOR, CONCRETE, PNEUMATIC

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers an internal, single probe, concrete pneumatic vibrator.

2.1 Government documents.

2.1.1 Specifications, standards and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

PPP-P-40	- Preservation and Packing of Hand Tools; Tools and Tool Accessories for Power Driven, Metal and Woodworking Machinery.

MILITARY

MIL-T-704	- Treatment and Painting of Materiel.
MIL-O-13417	- Oilers, Air Line, for Pneumatic Tools.
MIL-C-46168	- Coating, Aliphatic Polyurethane, Chemical Agent Resistant.

Beneficial comments (recommendations, additions, deletions) and any pertinent, data which may be of use in improving this document should be addressed to: USA Belvoir Research, Development, and Engineering Center, ATTN: STRBE-TSE, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at end of this document or by letter.

AMSC N/A FSC 3895

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STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129	- Marking for Shipment and Storage.
MIL-STD-130	Identification Marking of US Military Property.
MIL-STD-838	- Lubrication of Military Equipment.
MIL-STD-889	- Dissimilar Metals.
MIL-STD-1472	Human Engineering Design Criteria for Military Systems, Equipment, and Facilities.
MIL-STD-1474	- Noise Limits for Army Materiel.

HANDBOOKS

MIL-HDBK-113	Guide for the selection of Lubricants, Functional Fluids, Preservatives and Specialty Products for use in ground equipment systems.

(Unless otherwise indicated, copies of federal and military specifications, standards and handbooks are available from the Naval Publications and Form Center (ATTN: NPOAS) 5801 Tabor Avenue, Philadelphia, PA 19120-5099).

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 380 - Methods for Testing Rubber Hose.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

NATIONAL MOTOR FREIGHT TRAFFIC ASSOCIATION, INC., AGENT

National Motor Freight Classification

(Application for copies should be addressed to the American Trucking Association, In., ATTN: Traffic Order Section, 2200 Mill Road, Alexandria, VA 22314.)

UNIFORM CLASSIFICATION COMMITTEE, AGENT

Uniform Freight Classification Ruler.

(Application for copies should be addressed to the Uniform Classification Committee, ATTN: Tariff Publishing Officer, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

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SOCIETY OF AUTOMOTIVE ENGNEERS, INC. (SAE)

J1362 - Symbols for Controls, Indicators and Tell-Tales for Off-Road, Self-Propelled Work Machines Symbol 8.7.3.

(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description. The pneumatic concrete vibrator (hereinafter referred to as "vibrator") shall consist of a vane-type airmotor, and eccentric weight, a flexible hose handle, an air hose, an airline oiler, and an air strainer. The airmotor and the eccentric weight shall be encased within a waterproof cylinder to form the vibrating head and shall withstand continuous operation.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.3.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration prevention and control. The vibrators shall be fabricated from compatible materials, inherently corrosion resistant or treated to provide protection against the various forms of corrosion and deterioration that may be encountered in any of the applicable operating and storage that may be encountered in any of the applicable environments to which the vibrators may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals shall not be used in intimate contact with each other unless protected against galvanic corrosion. Dissimilar metals and methods of protection are defined and detailed in MIL-STD-889.

3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with component and subcomponent, and shall make information available upon request to the contracting officer or designated representative.

3.3.2 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the vibrator may be newly fabricated from recovered materials to the maximum extent practicable, provided the vibrator produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall be incorporated in the vibrator.

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3.4 Components.

3.4.1 Vibrating head. The vibrating head shall be made of steel and shall be not less than 13.00 inches (330 mm) and not more than 19.00 inches (483 mm) in length. The vibrating head shall be cylindrical and not more than 2.75 inches (70 mm) in diameter. The vibrating head shall have a waterproof housing for the airmotor and the eccentric weight. The total weight of the vibrating head, including the airmotor and the eccentric weight, shall be not less than 10.00 pounds (4.5 kg) and not more than 20.00 pounds (9 kg).

3.4.2 Eccentric weight and airmotor. The eccentric weight shall weigh not less than 2.50 pounds (1.125 kg) and shall be mounted on antifriction bearings. The airmotor shall be of the vane type and shall be directly connected to the eccentric weight.

3.4.3 Throttle. The vibrator shall be provided with a twist-type throttle valve for regulating the frequency of vibrations. The throttle shall remain in any set position between the closed and open positions. The throttle shall be labeled "SPEED" and shall be provided with a label conforming to SAE J1500 for continuously variable speed range. The throttle shall have features to exhaust the air from the air motor.

3.4.4 Hose handle and airhose. The vibrator shall be provided with a flexible hose handle not less than 15.00 feet (4,572 mm) in length. The air inlet hose between the throttle valve and the vibrating head shall be encased within the hose handle. The exhaust air from the airmotor shall pass through the hose handle one end of the hose shall be attached to the throttle and the opposite end shall be attached to the vibrating head.

3.4.4.1 Burst pressure. The hose handle and airhose shall withstand a pressure equal to four times the applicable operating pressure specified, in the applicable specification sheet (e.g., 90 psi operating pressure shall have a minimum burst pressure of 360 psi) without evidence of leakage, rupture, or detachment of a fitting (see 4.5.2.4.2).

3.4.5 Airline oiler and airline strainer. An airline oiler shall be in accordance with MIL-O-13417, 0.500-pint size, and an airline strainer shall be furnished. The strainer shall be of the type which can be cleaned without disconnecting the air supply line.

3.5 Lubrication.

3.5.1 Lubricant. The procedure for the selection of lubricants shall be in accordance with MIL-STD-838, section 5. Lubricants selected shall be in accordance with MIL-HDBK-113, chapter 2. When the specification of the lubricant selected includes a requirement for a Qualified Products Lists (QPL), the. lubricant supplied shall be from a source that is listed on the applicable QPL.

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3.5.2 Initial lubrication service. Unless otherwise preserved for packaging, the vibrator shall be serviced with military lubricants before delivery to the Government.

3.6 Performance.

3.6.1 Operating characteristics. The vibrator shall withstand operation online pressures up to 110.00 pounds per square inch (psi) and shall also withstand operation with the head completely submerged in concrete. The maximum operating speed shall be not less than 9, 000 vibrations per minute (vpm) and the operating speed shall be adjustable to any desired speed between 5,000 and 9,000 vpm. The air required to operate the vibrator at maximum speed shall not exceed 40.00 cubic feet per minute (cfm) at 90.00 psi. The vibrator shall produce a centrifugal force of not less than 1600 pounds (720 kg) when operated at 9,000 vpm.

3.7 Human factors engineering. The vibrator shall be in accordance with accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, general requirements (4), controls (5.4), labeling (5.5), design for maintainability (5.9), operational and maintenance ground/shipboard vehicles (5.12) and hazards and safety (5.13) of MIL-STD-1472, as applicable.

3.8 Noise limits. The noise produced by the vibrator shall be in with MIL-STD-1474, category D requirements, with the exception of MIL-STD-1474, 5.2, 5.3, and 5.4, when tested in accordance with 4.5.2.8. The provision of MIL-STD-1474, 4.3 and 4.4 shall apply if and only if MIL-STD-1474, 5.1.1.2 procedures have been pursued and documented to the satisfaction of the procuring activity and written permission to exceed the 85 dB(A) limit is obtained from the contracting officer. Hazard signs shall be in accordance with MIL-STD-1474, 4.3, except the sign will state "HEARING PROTECTION REQUIRED WITHIN (specify) FEET". The sign shall be readable at the distance indicated on the sign.

3.9 Identification marking. The vibrator shall be identified in accordance with MIL-STD-130. The information shall be roll stamped on the vibrating head or the contractor's nomenclature plate shall be attached to the vibrator.

3.10 Treatment and painting. The portions of the vibrator normally painted, internally and externally, shall be cleaned, treated, and painted in accordance with MIL-STD-704, type F or G, as applicable. Unless otherwise specified (see 6.2), top coat color shall be camouflage green 383 in accordance with MIL-C-46168.

3.11 Workmanship. The vibrator shall be free of defects such as galvanic action resulting from dissimilar metals, rust, cracks, and other defects that could impair its operation or serviceability.

3.11.1 Surfaces. All parts, components, and assemblies of the vibrator including castings, forgings, molded parts, starlings, bearings, seals, machined surfaces, and welded parts shall be clean and free of sand, dirt, fins, pits, sprues, scale, flux, and other harmful extraneous material. All exposed edges shall be rounded or beveled.

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3.12 Safety. All exposed edges shall be rounded or beveled. All exposed high temperature surfaces shall be guarded to prevent inadvertent contact by operating personnel.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Component and material inspection. The contractor is responsible for ensuring that components and materials used are manufactured, examined and tested in accordance with referenced specifications and standards as applicable.

4.2 Classification of inspections. Inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).
c. Inspection of packaging (see 4.6).

4.3 First article inspection.

4.3.1 Examination. The vibrator shall be examined as specified in 4.5.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The vibrator shall be tested as specified in 4.5.2 through 4.5.2.9. Failure of any test shall be cause for rejection.

4.4 Quality conformance inspection.

4.4.1 Sampling. Sampling of accessories for examination shall be in accordance with MIL-STD-105.

4.4.2 Examination. Samples selected as specified in 4.4.1 shall be examined for the defects specified in 4.5.1, as applicable.

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4.4.3 Test. Each vibrator shall be tested as specified in 4.5.2.6. Failure of the test shall be cause for rejection.

4.5 Inspection procedure.

4.5.1 Examination. The vibrator shall be examined for the following defects:

Major

101. Materials not as specified (see 3.3).
102. Material not resistant to corrosion and deterioration, or treated to be resistant to corrosion and deterioration for the applicable storage and operating environments (see 3.3.1).
103. Dissimilar metals, as defined in MIL-STD-889, are not effectively insulated from each other (see 3.3.1.1).
104. Contractor does not have documentation available for identification of material, material finisher or treatment (gee 3.3.1.2).
105. Used, rebuilt or remanufactured components, pieces or parts incorporated in the remolding set (see 3.3.2).
106. Dimensions and weight not as specified (see 3.4.1).
107. Vibrating head not as specified (see 3.4.1).
108. Eccentric weight or airmator not as specified (see 3.4.2).
109. Throttle riot as specified (see 3.4.3).
110. Hose handle and airhose not as specified (see 3.4.4 and 3.4.4.1).
111. Oiler and air strainer not as specified (see 3.4.5).
112. Lubricants not as specified (see 33.5).
113. Human factors engineering not as specified (see 3.7).
114. Safety not as specified (see 3.12).

Minor

201. Identification marking not as specified (see 3.9).
202. Treatment and painting not as specified (see 3.10).
203. Workmanship not as specified (see 3.11).

4.5.2 Tests.

4.5.2.1 Test conditions. The vibrator shall operate as specified herein, without maintenance, other than the contractor's recommended normal scheduled maintenance as established by a maintenance schedule prepared and submitted by the contractor prior to the test. During the test, the vibrator shall be lubricated with the oils and greases specified in 3.5. Oils shall be those designated for use in the ambient at the place of the test. Human factor considerations (see 3.7) shall be evaluated throughout testing.

a. During the tests failure of the vibrator the design or performance requirements specified herein, shall be caused for rejection.
b. Spillage or leakage of any fluid, including fuel, coolant, lubrication or hydraulic fluid, under any operation condition except as allowed herein.

4.5.2.2 Frequency of vibration and air consumption. Air used for this test shall have a line pressure of 90 psi. Install a flow meter in the airline between the air supply and the vibrating head and operate the

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vibrator at 9,000 vpm. Determine the line pressure, the air consumption in cfm, and the speed in vpm. An airline pressure other than 90 psi, air consumption of more than 40 cfm, or a speed of less than 9,000 vpm shall constitute failure of this test.

4.5.2.3 Minimum and intermediate speed. Using the hand throttle to control the speed, operate the vibrator at 5,000 vpm and at various speeds between 5,000 and 9,000 vpm. Inability to operate at any speed between 5,000 and 9,000 vpm shall constitute failure of this test.

4.5.2.4 Endurance. Completely submerge the vibrator head in a simulated concrete mixture consisting of gravel, sand, clay, and water. Operate the vibrator for not less than 100 hours continuously with a line pressure of not less than 90 psi and at a speed of not less than 9,000 vpm. Determine and record the speed of the vibrator in vpm at intervals of 30 minutes throughout the test. Inability of the vibrator to operate for 100 hours continuously or at a speed of less than 9,000 vpm shall constitute failure of this test.

4.5.2.4.1 High pressure. During the last 30-minute period of the test specified in 4.5.2.4, operate the vibrator with a line pressure of not less than 110 psi. Air leaks or damage to any part of vibrator shall constitute failure of this test.

4.5.2.4.2 Burst pressure. The burst pressure test of the hose handle shall be in accordance with ASTM D 380. The test pressure shall be equal to or greater than four times the maximum operating pressure as specified in 3.4.4.1. Evidence of leakage, rapture, or detachment of a fitting shall constitute failure of this test.

4.5.2.5 Air strainer. During the test specified in 4.5.2.4, remove and clean the airline strainer. Inability to remove and clean the airline strainer without disconnecting the airhose shall constitute failure of this test.

4.5.2.6 Operational. Operate each vibrator for not less than 5 minutes. Inability to start the vibrator, or to control the vibrator speed with the throttle, or inability of the vibrator to operate continuously for 5 minutes shall constitute failure of this test.

4.5.2.7 Centrifugal force. Determine the centrifugal force produced by the vibrator. The force shall be determined either by an accelerometer or the contractor may furnish calculations as proof of the force developed by the vibrator. A centrifugal force of less than 1600 pounds (720 kg) at 9,000 vpm shall constitute failure of this test.

4.5.2.8 Noise-level. Noise levels shall be measured in accordance with MIL-STD-1474, Category D requirements and reported in type format indicated by MIL-STD-1474, figure 7. As a minimum, noise levels shall be measured when equipment is operating under full load. MIL-STD-1474, 5.1.2.1.4 contours shall be taken at not fewer than 12 equal (horizontal) arc increments; one increment shall include data from the noisiest position. Additionally, the noise level at the typical operating position shall be provided as dB(A) level. Failure to comply with MIL-STD-1474 provisions shall constitute failure of this test.

4.5.2.9 Transport. Transport the vibrator in a construction type vehicle, such as a 5-ton size dump truck, for a total distance of not less than 75 miles, at an average speed of not less than 25 miles per hour.

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The vibrator shall not be crated or packaged and the vehicle shall not be loaded with other material. The road for this test shall have a rough surface similar to the roads encountered on construction sites. After completion of the transport test, examine the vibrator for damage; operate the vibrator for a period of time sufficient to demonstrate that the vibrator was not damaged during transport. Evidence of any damage, or failure of the vibrator to operate satisfactorily shall constitute failure of this test.

4.6 Inspection of packaging. The preservation, packing, and marking for level A, B, and C shall be examined to determine compliance with the applicable quality assurance provisions of PPP-P-40 and any additional requirements in section 5.

5. PACKAGING

5.1 Preservation. Preservation shall be level A.

5.1.1 Level A. The complete vibrators shall be preserved and packaged in accordance with level A requirements of PPP-P-40.

5.2 Packing. Packing of the vibrators shall be level A, B or C as specified (see 6.2).

5.2.1 Level A and B. The vibrators, preserved as specified in 5.1, shall be packed in accordance with level A or B requirements of PPP-P-40.

5.2.2 Level C. The vibrators, preserved as specified in 5.1, shall be packed in accordance with the Uniform Freight classification Rules or the National Motor Freight Classification Rules.

5.3 Marking. Marking shall be in accordance with MIL-STD-129. Level C marking shall be in accordance with the Uniform Freight Classification Rules or the National Freight Classification Rules. Each shipping container shall be marked with the cube and gross weight.

6. NOTES

(This section contains information of a general or explanatory nature which is helpful, but is not mandatory.)

6.1 Intended use. The vibrator is intended for use in consolidating low-slump, freshly pawed, Portland cement concrete.

6.2 Acquisition requirements. Acquisition documents shall specify the following:

a. Title, number, and date of this specification.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.11 and 2.2).
c. When a first article is required for inspection and approval, and the number of units required (see 3.2).
d. When treatment and painting shall be other than specified (see 3.9).
e. Color required when other than specified (see 3.9).

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f. Level of Packaging required (see 5.2).

6.3 First article. When a first article inspection is required, the item should be a preproduction model. The flat article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests, and approval of the first article test results and disposition of the first article. Invitation for bids should provide that the Government reserves the right to waive the requirement for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so, in the solicitation.

6.4 Lubricants. MIL-STD-838 prescribes the policy for using specification type products wherever possible and provides specific requirements for potential use of non-standard proprietary products. MIL-STD-838 is implemented by MIL-HDBK-113. The contracting officer should note that unless otherwise authorized by the US Any Belvoir Research, Development and Engineering Center (ATTN: STRBE-VF), Fort Belvoir, VA 22060-5606, lubricants, fluids, and greases for ground equipment systems must be restricted to those listed under MIL-HDBK-113, chapter 2.

6.5 Technical manuals. The requirement for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on a separate contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.6 Provisioning. The contracting officer shall include provisioning requirements for repair parts, maintenance tools, operational instructions and accessories as necessary (including any special tools), and instructions on shipment of the vibrator(s).

6.7 Subject term (key word) listings.

Concrete, pneumatic, vibrator
Concrete, vibrator, pneumatic
Pneumatic, concrete vibrator
Vibrator
Vibrator, concrete, pneumatic

6.8 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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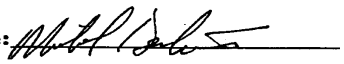
6.9 AQL certification. This specification is certified to be in compliance with current Army Materiel Command (AMC) policy for the elimination of AQL's/LPTD's (Acceptable Quality Levels/Lot Tolerance Percent Defectives) from military specifications.

(Certified by/date: 

Custodians
Army – ME
Air Force – 99

Preparing Activity
Army – ME

Review activities:
Air Force – 84
DLA – CS

Project 3895-0290

User activities:
Army – AT
Navy – MC, YD

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